

Alireza Rezapour

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Backend engineer with 3+ years owning services from monolith-split to event-driven scale, plus production LLM/RAG work. Comfortable across Node.js, Go, Python and the messy edges in between (PostgreSQL tuning, Kong plugins, hybrid retrieval). Shipped wins: 90% fewer failed transactions, 99% read-load reduction, 70% MTTR cut, hallucination under 3% on a UK-lender RAG product.

EXPERIENCE

Hamtapay

Iran, Mashhad — *Backend Engineer*

Aug 2025 - Present

- Refactored payment-and-transfer engine core: tightened transaction isolation, optimized hot-path queries with targeted indexes, and tuned PgBouncer pooling — cutting transaction failures 90% and production incidents 30%.
- Introduced read-side caching on hottest endpoints, dropping DB read load 99% (from 5,000+ req/s to under 20 req/s).
- Migrated synchronous inter-service calls to RabbitMQ-based event-driven messaging, cutting cross-service network traffic 40% and decoupling failure domains.
- Migrated analytical workloads from PostgreSQL to ClickHouse, halving dashboard p95 latency and offloading ~50% of PostgreSQL load.
- Extended Kong API Gateway with custom Go plugins for auth, rate-limiting, and request shaping across production traffic.
- Drove team-wide AI-assisted development practices (Cursor .rules, Claude CLAUDE.md, prompt patterns), cutting dev cycle time ~20% and lowering LLM token spend through better tool and agent usage.
- Took ownership of legacy "What to Buy" top-4 crypto recommendation service: replaced underlying prediction LLM, added validation layer to filter low-confidence picks, improving recommendation quality.

Criterion (Sub-brand) — *Lead Engineer*

- Built Criterion, an agentic RAG assistant over 50+ UK-lender lending-criteria documents with multi-turn context memory, follow-up question generation, and citation-grounded answers; replaced pure-embedding retrieval with AI-labeled chunk lookup over SQLite to fix recall on small corpora, holding LLM hallucination rate under 3% via strict prompting and citation grounding, with sub-60s response latency.
- Shipped fully AI-driven daily blog auto-publishing one lending-criteria news article per day with generated imagery and cited sources. Stack: LangGraph, LangChain, Python, FastAPI, OpenAI / OpenRouter.

SKILLS

Language

Javascript/Typescript, Go, Python

Database

MongoDB, PostgreSQL, Redis, ClickHouse, InfluxDB

Cloud & DevOps

CI/CD (GitLab, GitHub), Docker, Podman, API Gateway (Kong, KrakenD), S3, Nginx

Frameworks

Nest.js, Express.js, Adonis.js, FastAPI

Messaging

RabbitMQ, NATS, Kafka

Architecture & Practices

Microservices, Event-Driven, Hexagonal, System Design, TDD, Cache Strategies, REST, gRPC, Clean Code

AI/LLM

RAG, Agentic Workflows, LangChain, LangGraph, Vector Search, Embeddings, LLM Prompting, OpenAI / OpenRouter

Methodologies

Agile/Scrum, Jira, Mentoring, Code Review

Part Software Group

Iran, Mashhad — *Backend Engineer*

Apr 2024 - Aug 2025

- Resolved B2B-service outages (payments, auth, cron) via cross-team response and post-mortem ownership to prevent recurrence.
- Migrated ~12 backend services from pm2-based deploys to a new internal Kubernetes-like platform, cutting deploy time from ~1 working day to minutes; onboarded team to the new workflow.
- Designed and built B2B infrastructure services (payment, user, auth, cron) using hexagonal architecture and event-driven patterns to keep core domain logic decoupled and testable.
- Diagnosed peak-time socket exhaustion across services, traced root cause to synchronous HTTP fan-out from cron jobs; authored proposal for message-queue architecture later adopted org-wide.
- Authored RFCs on logging/observability, error-handling and API contracts, and service boundaries; standards adopted across multiple teams to reduce integration friction.
- Acted as primary incident point of contact, leading critical-outage calls and cross-team meetings on behalf of tech lead.
- Mentored junior engineers via code reviews, pair programming, and architectural guidance, raising team velocity and code-review quality.

Forvest

Iran, Mashhad — *Backend Developer*

Aug 2023 - Apr 2024

- Contributed to monolith-to-microservices migration across core services, splitting domains and shortening deploy cycles.
- Redesigned ~200-symbol crypto market-data ingestion service for higher fetch throughput, unblocking real-time AI analytics.
- Adopted NATS for low-latency inter-service streaming, feeding downstream AI/analytics consumers.
- Deployed KrakenD API Gateway for routing, rate-limiting, and horizontal scaling of backend endpoints.

Farajahan Hashtom

Iran, Mashhad — *Backend Developer*

Jan 2023 - Aug 2023

- Built Node.js/Express REST endpoints with MongoDB persistence on backend team.
- Implemented real-time WebSocket price-broadcast feature for live trading data.
- Integrated BEP20/ERC20 wallet flow: per-user wallets on signup, on-chain balance polling via cron, deposit reconciliation to DB.

PUBLICATIONS

medium.com/@namatery

EDUCATION

Bachelor's Degree in Computer Engineering

PROJECTS

[multer-minio](#) - TypeScript storage engine for Multer enabling file uploads to MinIO; published to npm with 6 GitHub stars and external forks.

[blackbird](#) - Self-built reverse proxy in TypeScript with built-in Node cluster and load balancing.

[BlockMail](#) - Decentralized email PoC: encrypted messages between Ethereum addresses via IPFS storage and on-chain events; built with Electron + Hardhat.

[xo-villain-ai](#) - Tic-tac-toe with LLM-powered trash-talking opponent: Vue UI + FastAPI backend (minimax + LLM banter) with SQLite persistence.

OPEN SOURCE

- [Uptime Kuma](#) — merged feature PR (customizable Matrix notifications)
- [Firecrawl](#) — open PR (test coverage for HTML srcset handling)
- [podman-mcp-server](#) — 2 merged PRs (JSON output for CLI; AI-assistant docs)